Final Project Guidelines: Reservoir Engineering 2

I. Overview

This final project aims to evaluate students' ability to apply theoretical knowledge in Reservoir Engineering by developing a Python-based analytical tool. The project emphasizes transforming engineering concepts into a functional digital solution capable of solving real-world reservoir problems. The project is divided into two phases: (1.) Selection and Preparation Phase, (2.) Testing and Presentation Phase

II. Phase 1: Selection and Preparation

1. Group Formation

Each group must consist of three (3) members.

2. Topic Selection

Each group must choose one (1) of the following topics:

- Volumetric Calculation / Monte Carlo Simulation
- Bulk Volume Calculations
- PVT Analysis
- Material Balance Equation (MBE)

3. Tool Development

Each group must develop a Python-based tool aligned with their selected topic. The tool should be capable of:

- Accepting input data (manual or file-based)
- Performing necessary engineering calculations
- Producing interpretable outputs (tables, graphs, or summaries)

4. Expected Outputs per Topic

Topic	**Expected Output**
Volumetric Calculation / Monte Carlo Simulation	Reserves estimation data and interpretation
Bulk Volume Calculations	Accurate bulk volume computed results
PVT Analysis	Fluid classification and analyzed fluid properties
Material Balance Equation (MBE)	Reservoir Performance Data and Its Drive Mechanism Analysis

5. Tool Requirements

- The Python tool must:
- Be functional and user-friendly
- Handle real-world data inputs
- Include clear computational logic
- Provide interpretable results (graphs, tables, or insights)
- Be efficient in execution time
- Include basic validation (error handling for invalid inputs)

III. Phase 2: Testing and Presentation

1. Testing

Actual reservoir data will be provided. Each group must:

- Input the data into their tool
- Generate results
- Perform analysis based on outputs

2. Submission Requirements

- Final output results
- Interpretation/analysis of results
- Code file or executable tool

3. Grading System

A. Accuracy-Based Score:
   - Correctness of calculations
   - Reliability of results
   - Quality of interpretation and analysis

B. Time-Based Score:
   - Speed of execution
   - Efficiency in generating and analyzing results

Tips:

The tool developed in Phase 1 will be used directly in Phase 2, so ensure it is tested, optimized, and ready for real data application.

Groups are encouraged to:
- Use visualization tools (graphs, plots)
- Keep interface simple but effective
- Ensure code readability and organization